

Tree-Based Machine Learning Regression Models for Predicting Marathon Race Times

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Abstract—Long-distance endurance running has recently increased in popularity, and the integration of technology has become a staple within the running community. Marathon runners are relying more on predictive algorithms in their tracking devices to anticipate their race-day performance, yet such algorithm development remains a complex challenge. Therefore, this study aims to evaluate tree-based machine learning regression models for predicting an athlete's marathon race time and identify a superior model among Random Forest and XGBoost. Multiple progressive experiments were conducted under a controlled environment while utilizing different aggregated training block periods for a specific demographic, focusing on their effect on the predictive performance. Results demonstrated that the Random Forest regressor remarkably surpassed the predictive capacity of XGBoost throughout all time steps, with the model reaching its peak predictive performance at Week 6. Such results highlight the importance of acute training data and the strength of predictions by simple tree-based machine learning models in endurance running. This study also suggests possible areas for future research in this domain to bolster the running community with improved predictive models and target other demographic groups.

Index Terms—Marathon Race Time Predictions, Tree-Based Machine Learning Regressors, Random Forest, XGBoost, Training Block Data

I. INTRODUCTION

In recent years, various health benefits have been associated with regular physical activity [1]. As a result, long-distance endurance sports such as marathons, ultra-marathons, and triathlons have increased in popularity [2] – [4]. Esteve-Lanao et al. [2] state that participation in marathon events by recreational runners has become immensely popular worldwide. Additionally, wearable tracking devices have become a common necessity [5][6] as they provide real-time sport-related data (e.g., pace, heart rate). This, in turn, helps individuals achieve a target performance, either during training or a race.

Predicting marathon race time (MRT) is highly important and provides various entities with valuable insights, but some researchers [2][5][7] have considered it a difficult task. Among others, the underlying reasons point toward the complexity of the sport itself, including the event landscape, unpredictable environmental conditions, different runner biometrics [7], and different tracking devices [5]. This led to the introduction of machine learning (ML) to improve accuracy, including tree-

based algorithms (supervised), clustering (unsupervised), and time-series algorithms.

The purpose of this experimental study [8] is to investigate the use of tree-based ML models to predict an athlete's MRT and subsequently evaluate the comparative predictive performance of Random Forest (RF) and Extreme Gradient Boosting (XGBoost) regressors across diverse historical data. The dependent variable of this study is the predictive performance of the ML algorithms in estimating the continuous MRT value, while the independent variables will be the dataset and the model type (RF against XGBoost). Specifically, historical training data will be aggregated into blocks starting from 16 weeks, iteratively reduced by 2-week intervals, until reaching a minimum of 6 weeks. Moreover, the control variables will be the dataset features, including activity metrics and demographic information, and the repeated k-fold cross-validation technique, so that each ML regressor can make decisions on the same data. This research does not take into consideration the landscape or weather conditions, nor any anthropometric or physiological factors that might alter the result.

To achieve this, the study aims to explore the effectiveness of tree-based ML regressor models in predicting an athlete's MRT and to identify the optimal model based on predictive performance. Furthermore, this study seeks to determine whether the number of weeks of historical data yields a quantifiable increase in the models' predictive performance. Specifically, the hypothesis of this research states that an XGBoost ML regressor will achieve higher predictive performance than an RF regressor. Additionally, it is expected that XGBoost will be most accurate when trained on larger volumes of historical activity data (16-week training block), as this will supply the model with more comprehensive data about an individual's marathon buildup.

In view of the previously mentioned factors and as illustrated in Fig. 1, a positivist philosophy will be adopted as this study is highly focused on gathering quantifiable outcomes (predictive performance). A deductive approach will be implemented because a hypothesis has been established, and the study is designed to prove or disprove it.

In addition, an experimental strategy will be conducted, given that the research has a dependent variable (accuracy), two independent variables (model type and time frame of

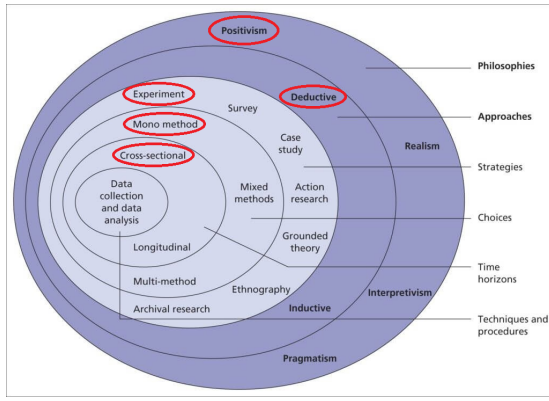


Fig. 1. The Research Onion - Adopted by [2]

historical data), and various control variables to ensure an unbiased investigation. Thus, a mono-method quantitative research choice will be adhered to, as no surveys, interviews, or questionnaires will be held. It will depend exclusively on an open-source dataset as input and values obtained from different experiments as output. Consequently, although the dataset encompasses a period of time, a cross-sectional time horizon will be utilized. This is because the experiments will leverage a snapshot of aggregated data, rather than monitoring athletes during a specific training block.

II. LITERATURE REVIEW

The running community has grown immensely over the past few years, driving the demand for wearable tracking devices that capture real-time, sport-related data. Due to the complexity of forecasting marathon race times, predictive machine learning (ML) algorithms have become a prevalent and necessary resource for individuals in training and attaining target performances. As this technological integration has transformed how athletes strategize for major marathon races, various researchers have acknowledged and sought to improve these predictive models.

This section evaluates research methodologies adopted from a chosen set of studies and analyses the different approaches and frameworks employed by the researchers. Fig. 2 illustrates the selected set of academic studies, presenting a more comprehensive understanding of the contributions and interconnections to the predictive side of this domain. To deliver a systematic and comparative analysis, this section is based on the research onion model proposed in [9] and the research design framework proposed in [8]. Common methodological patterns will be highlighted, along with their advantages and disadvantages.

A. Instance-Based, Regression, and Deep Learning Models for Performance Prediction

With very similar objectives in mind, all six studies [3] [4] [5] [6] [7] adopted a positivist standpoint, considering an individual's performance in a marathon event a continuous, measurable, and objective outcome that can be precisely

calculated by machine learning (ML) techniques. Hence, they were all distinctly deductive in nature as they validated existing hypotheses using experimental designs and historical data. Almost all studies utilized longitudinal data gathered over multiple years, except for [4]. Additionally, they entirely depend on quantitative, numerical data, distinguishing them as mono-method quantitative methodological studies. No qualitative information was gathered about an athlete's emotional state or training experience.

Both Feely *et al.* [3] and Lerebourg *et al.* [4] employed the same instance-based learning approach as a regressor to predict athlete marathon race times (MRT). The difference was that the former study coupled the k-Nearest Neighbour (KNN) model with a case-based reasoning (CBR) system, which helped bypass the necessity of complex physiological domain models and mitigated the likelihood of unrepresentative training recommendations. The latter study compared a KNN model against an Artificial Neural Network (ANN), on the same objective. A clear difference is observed in the time horizon when choosing longitudinal [3] and cross-sectional [4] historical datasets.

Feely *et al.* [3] gathered thousands of activity records from 19,930 individuals between 2014 and 2017, and notably, due to forecasting difficulties, the runners were eligible based on their marathon performance. It included finish times between three and five hours for females and less than five hours for males, and although this resulted in better accuracy (even through visual illustrations), including sub-three-hour female athletes should have been considered, as it provides variance. On the other hand, Lerebourg *et al.* [4] investigated solely using four static inputs from 820 runners, including at least one 10-kilometer race result. These were appropriately normalized before being fed to the ANN model, aligning with the sigmoid transfer function, and to eliminate scale variations between the input variables for the KNN model, for accurate Euclidean distance calculations.

Regarding this study [3], the large-scale dataset was essential and proved highly beneficial when performing 3 separate experiments on feature variations. Although the Root Mean Square Error (RMSE) showed the models' improved performance over time, one must consider that it was not put through real-world trials. Furthermore, the dataset by Lerebourg *et al.* [4] was effective in addressing overfitting and showed that KNN outperformed the ANN model; however, incorporating a half-marathon race result from the same year could improve the model's predictive performance. This study [3] also stated that the k value was set to 15 after showing a point of error stabilization, whilst Lerebourg *et al.* [4], although having an acceptable Mean Absolute Error (MAE), set the k value to 3 without any explicit reasons. This could mean that other values could have led to better results. A common shortcoming of both studies is gender imbalance, with a 24% [3] and 17% [4] female sample population, which could result in inaccurate training recommendations and MRT prediction for this gender.

In another study by Ruiz-Mayo *et al.* [5], two regressors were evaluated and compared with a baseline ZeroR model:

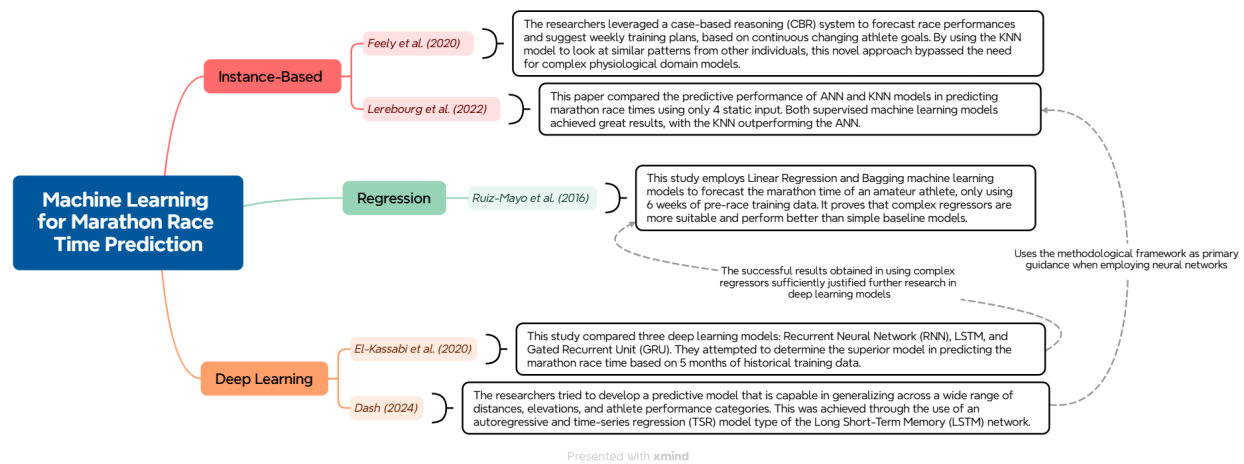


Fig. 2. Literature Map

Linear Regression (LR) and Bagging, to assess the validity of more complex regressors for forecasting the MRT. Starting with the time horizon, the 6-week longitudinal dataset selection contradicts the researchers. It was explicitly mentioned that several months of preparation are required to compete. Without any explanation for this period selection, it might depict uninformed decisions, as this hides the majority of the training period, which could have resulted in better model performance. Adequate filtering was held, and relevant features were engineered based on weekly, bi-weekly, and the full training window. These supply the models with significant information, which contributed towards low MAE values, showing the importance of rigorous pre-processing.

The issue with the employed dataset was the population size, which contained only 1,000 suitable athletes from the original 3,386, coming from two major marathon events in the first half of the year, whilst there are a total of six. Therefore, generalizability could be difficult when applied to others. Moreover, using the Weka and Knime software and the standard 10-fold cross-validation, the researchers [5] developed a Bagging model of 100 combined regression trees that attained the lowest MAE of all the models. Thus, testing the algorithm with slightly more regression trees at a time could reveal better results. Nonetheless, the study still meticulously held the three separate experiments, starting with the original, filtered, and then the injury-free dataset.

Furthermore, both Dash *et al.* [6] and El-Kassabi *et al.* [7] leverage DL models for their investigations, as the former employed a regression and time-series Long Short-Term Memory (LSTM) model, while the latter compared the predictive performance among a Recurrent Neural Network (RNN), LSTM, and Gated Recurrent Unit (GRU). Although DL models were used, the objective differed. El-Kassabi *et al.* [7] focused on predicting the MRT, and Dash *et al.* [6] developed a model that forecasted an athlete's performance across a wide range of distances, elevations, and performance categories. Both studies based their experiments on longitudinal and sequential historical training logs, with an average of 6.6 years [6] and 5

months [7]. The sequential aspect is of great importance and was rightly considered due to the nature of DL models. Also, the time horizon aspect provided the algorithms with valuable information, which was reflected in the results.

Regarding the participants, Dash *et al.* [6] had one issue, as only 15 recreational runners were included in the study. Additionally, one model was developed and modified for each individual. Thus, future work would suggest gathering and testing the superior model on a larger sample population and developing a generalized LSTM predictive model, as this is not contextually feasible. This would result in more training patterns, enhancing model training, increasing scalability, and highlighting certain flaws and possible improvements. However, it must be noted that the specialised filtering process exhibited lower Mean Absolute Percentage Error (MAPE) values, showing its significance again, as in [5]. Even the hyperparameter optimization framework (Optuna) was selected strategically, likely due to its efficiency. Overfitting was also addressed by having two regularization methods, dropout and early stopping.

Analysing the dataset in this study [7], the preparation period was more than accounted for, as mentioned in [5], and the experiments were systematically designed and executed, starting with 25 and then 50 hidden states for each model. To further enhance the dataset, the statistical three-sigma rule was adopted to eliminate outliers, yielding a highly concentrated dataset. This ensured an unbiased comparison and clear evaluation of the superior GRU model, primarily through the MAPE metric, as in [6]. A possible issue would be generalising the model, similar to the issue in [5], since it was only applied to the Boston marathon. Thus, this should be tested on other major marathons to validate its applicability. The experimental findings also showed that more hidden states contributed to slightly more precise predictions; therefore, future research should assess how MRT predictions change with gradual increments in hidden states.

III. RESEARCH METHODOLOGY

After analyzing the methodologies outlined in Section II, it was noted that researchers in numerous studies ([10], [11], [12], [13], [14]) employed a comparative approach, which evaluated machine learning (ML) models from the same or different paradigms. These investigations were conducted to determine whether more complex algorithms are necessary to achieve a common objective, concerning prediction accuracy in long-distance running events. In keeping with these methodological approaches, our study endorsed a similar strategy, performing progressive and comparative experiments aimed at addressing the following research questions:

- **RQ1:** Which tree-based machine learning regressor obtains the highest predictive performance when forecasting an athlete's marathon race time?
- **RQ2:** To what extent does the amount of weekly historical data affect the predictive accuracy?

All six individual experiments are conducted to collectively address RQ1 and RQ2, in which two ML regressors are evaluated based on their predictive performance on progressively reducing data, coupled with a repeated k-fold cross-validation technique for each experiment. In this manner, a superior model is established. This will help analyse the effects of different weekly historical data from different evaluation metrics and determine whether a model remains consistently better. To address the research questions, this paper presents the following objectives, which are also outlined within the research pipeline (Fig. 2):

- 1) Obtain a dataset containing daily activity records from individuals and filter to the dominant demographic (gender and age group).
- 2) Check for major marathon events that have at least 16 weeks of daily training data.
- 3) Aggregate the training block data of each athlete into a single record while engineering new features.
- 4) Conduct benchmark testing using the reference solution [15] and its utilised dataset.
- 5) Test and evaluate the predictive performance of both ML regressors on progressively reducing data to identify which model adjusts best to different data availability.

A. Methodology Approach

This approach supports the positivist philosophy and deductive approach conveyed in Section I, which highlights a systematic and rigorous analysis of ML regressors for predicting an individual's marathon race time (MRT), while gathering measurable outcomes through their predictive performance. During each experimental course, a cross-sectional dataset will be supplied to each model, independent of the historical timeframe being assessed. Therefore, a mono-method quantitative research choice will be followed. Consequently, our experiments align with the Research Onion model illustrated in Fig. 1, maintaining consistency in our research methodology.

B. Research Pipeline and Methods – Experiment Design

The research pipeline in Fig. 3 outlines the main phases required to achieve the stated objectives and answers the research questions posed in this paper. The foundational work necessary for the experiments (Phases 01 and 02) was conducted in a similar study that aimed to predict the MRT using sequential deep learning (DL) models, and a reference solution was also provided [15]. In turn, the desired dataset was provided with the target value. This was also highly specific to the dominant demographic group within the dataset (males aged 35 to 54 years). Additionally, from the six available marathon majors, the training block concerning the Tokyo 2019 marathon was removed as it contained only 8.5 weeks of data.

In Phase 03, this data source was utilised, and additional features were engineered from two native input features: distance (kilometres) and duration (minutes). The individual training blocks were then combined into a single record per athlete, and a repeated k-fold cross-validation technique was applied to the data files for model training and testing within Phase 04. Therefore, the last phase focused on obtaining quantitative results by employing and comparing two ML regressors: Random Forest (RF) and XGBoost, starting with the data-rich train-test files and progressively utilising those that had less weekly training data.

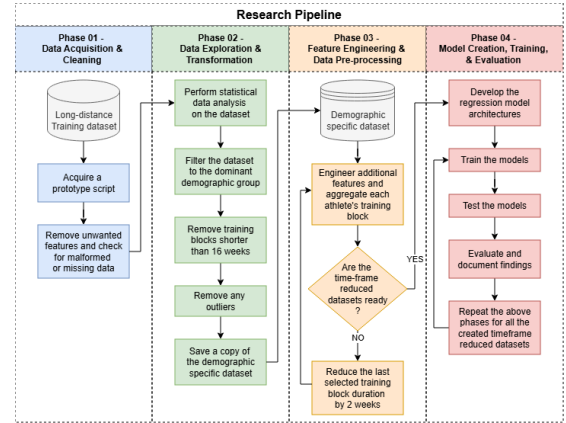


Fig. 3. Research Pipeline

RQ1: Which tree-based machine learning regressor obtains the highest predictive performance when forecasting an athlete's marathon race time? and **RQ2:** To what extent does the amount of weekly historical data affect the predictive accuracy?

This subsection addresses both research questions concurrently; the progressive data-reduction experiments designed for RQ2 ultimately generate the performance metrics required to identify the superior tree-based model for RQ1. This also provides further analysis on how both models adjust and perform when provided with fewer weeks of training data.

Primarily, benchmark testing was held using the exact reference solution code and aggregated data [15] on Linear

and Random Forest (RF) regressors. Since XGBoost is also investigated in this research, this model was implemented within the same reference code [15] to maintain consistency. Thus, three benchmark models were developed to enable a fair comparison of results. Due to the unpredictable and irregular nature of physical exertion by an athlete, the Linear regressor was not considered for this research.

The following experiments were conducted using two ML regressors, RF and XGBoost, based on three criteria: (a) similar underlying algorithmic structures, (b) common data input ratios, and (c) no feature scaling required. Therefore, this contributed to a better comparative study, specifically analysing the effectiveness of tree-based ensemble regression models. A sequentially ordered activity dataset, consisting of 16-week training blocks per athlete from the dominant demographic, was progressively reduced in 2-week intervals until reaching 6 weeks and hence aggregated. Another study [12] also used tree-based ensemble regression models, but only with 6 weeks of data; thus, this led to such a methodological approach. Some inessential features were removed, such as rolling metrics and weekly metrics, and new ones were formulated from the original distance and duration column values. This ensured that the models were supplied with essential information to aid performance predictions, as previously done by [10]. Notably, the original dataset [16] was open-source and obtained from Kaggle.

During training, model architectures were kept the same as those developed within the benchmark models to maintain coherence. The models were evaluated using standard 10-fold cross-validation, meaning the data was iteratively split into a 90:10 train-test ratio. This methodological step ensured a strong evaluation and mimics the approach taken in previous work [12]. Reliability was also established by repeating the experiment three times, thereby reducing the likelihood of random errors, an approach similarly adopted by [13]. Moreover, shared hyperparameters between the ML models were kept constant (such as the number of estimators) to ensure a fair structural comparison. The effectiveness of their predictive performance was assessed using the universal Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the R-squared value (R2) metrics, as proposed by [10] [11] [12] [14], and these are defined as follows:

$$MAE = \sum_{t=1}^n \frac{|y_{pt} - y_t|}{n} \quad (1)$$

$$RMSE = \sqrt{\frac{\sum_{t=1}^n (y_{pt} - y_t)^2}{n}} \quad (2)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (3)$$

To answer both research questions, the MAE and R2 values were key to deducing which model accomplished the closest prediction to the actual target MRT value, with the RMSE highlighting any limitations of the individual models. Each metric was accompanied by a standard deviation, given that

3 experiments were held on each dataset and model. These metrics were calculated using their respective imports from sklearn.metrics library.

C. Validity, Reliability, and Generalizability

Validity: Numerous steps were taken to ensure the validity of each experiment, including the domain-standard 10-fold cross-validation technique for all datasets, even after reducing some historical data, which inherently uses a 90:10 split ratio. Consequently, the same train and test data files were supplied to each ML regressor to ensure valid comparisons. The models were also chosen with intent, as they fall within the same ML category types and require no feature scaling. Construct validity was adhered to when developing new features, as they represented valid athlete physical exertions during the training block. Additionally, the evaluation metrics (MAE, RMSE, R2) were valid as they were used in similar studies [10] [11] [12] [14], and they provided complementary strengths to ensure a robust and valid evaluation.

Reliability: Having the model architectures and common hyperparameter values as control variables ensured experimental reliability. Using the domain-standard 10-fold cross-validation technique ensured stable and consistent results during training and testing. Moreover, both models underwent a repeated 10-fold cross-validation process (three times) for each data iteration to mitigate any random errors caused by data splitting, and the same technologies and environments were used. Notably, a common script was utilised when calculating and developing new features for all datasets.

Generalizability: Regarding this aspect, the results are partially limited because the study focused on a specific demographic, applying only to male runners aged 35 to 54. Therefore, employing the developed models on athletes from different demographic groups would yield inaccurate results. Also, only two ML regressors were used with a specific, consistent structure throughout. On the other hand, when the models were provided with unseen test data, they adjusted and performed predictions accordingly, and training data was gathered before five of the six major marathons. Consequently, making them applicable for most marathon majors for this specific demographic group. While the research provides valuable insight into MRT predictions, attention should be given when applied to broader contexts.

D. Technologies and Environment Used

The experiments were executed on a local machine with a 13th Gen Intel® Core™ i7-1355U CPU, 16 GB of RAM, and an integrated Intel® UHD Graphics GPU, drawing from 7.8 GB of the system RAM. Since the ML regression models did not require high computational power for training and testing, this setup was more than enough. All the prototypes were developed and executed in a dedicated Python 3.11.14 environment using a Jupyter Notebook with multiple sections, as it provides a lot of flexibility and compatibility. Additionally, key dependencies and libraries included NumPy, Pandas,

Matplotlib, Seaborn, Scikit-learn, Statsmodels, TensorFlow, and Dill, managed via the conda-forge channel.

E. Ethical Considerations

While conducting this study, an open-source dataset was used, consisting of running activity gathered from public athlete Strava pages. As a data source, it was used in numerous studies. It is licensed under the Creative Commons Attribution 4.0 (CC BY 4.0, Deed – Attribution 4.0 International – Creative Commons) licence, and the data collection process was in accordance with the Brazilian Ethics Research Committee standards.

The individuals were anonymised, and no personal information was kept or shared to safeguard any sensitive information and the runners' identities. Only essential training session data was used, such as training distance and duration, gender, age group, country, and a list of previously run marathon majors. No harm was inflicted on the runners themselves as they were not instructed or forced to run for the benefit of this study; rather, they were primarily randomly chosen from Strava based on their participation in marathon majors in 2019 and 2020.

In mentioning Strava, a popular training application worldwide, the tracking devices used during the training sessions, their software, and those linked to them were not recorded as they did not form part of this study. Our study is purely aimed at predicting an athlete's running performances within different marathons while leveraging separate training efforts.

F. Generative Artificial Intelligence (AI) Considerations

Artificial Intelligence tools, specifically Gemini 3.1 Pro and Grammarly, were utilised to support the development of this paper, mainly for reconstructing the text for academic soundness, suggesting phrases and synonyms, and code snippets for the creation of the prototype. The methodological approach and decisions taken during the study were made entirely by the author, who fully takes authorship of the presented content.

IV. FINDINGS & DISCUSSION OF RESULTS

RQ1: Which tree-based machine learning regressor obtains the highest predictive performance when forecasting an athlete's marathon race time? and RQ2: To what extent does the amount of weekly historical data affect the predictive accuracy?

As acknowledged within Section III, we identified a superior Machine Learning (ML) regressor through numerous progressive experiments while utilising timeframe-reduced datasets. This also provides an exploratory analysis of the effects of different volumes of aggregated data. As previously mentioned, we used two key metrics, Mean Absolute Error (MAE) and R-squared (R^2), and an additional Root Mean Squared Error (RMSE) metric to quantify the results and answer the proposed questions.

The MAE value describes the average absolute distance, in minutes, between the predicted marathon race times (MRTs), even though they might have been faster or slower. Thus, all the values are considered as linear. Unlike the MAE, the R^2 ,

which is a unitless scale ranging from 0.0 to 1.0, is a measure showing how much of the MRT values it manages to comprehend to build relationships between the input features and the actual race day performance. On the other hand, the RMSE, which shows the amount of error in minutes, considers larger errors in prediction and shows whether a model is making gross predictive errors within the sample population. Table I shows the results obtained from the benchmark models from the reference solution, using the same metrics, while Table II represents a summary of the mentioned metrics for each model on different weeks of aggregated data for this study. Three visual representations, showing how the values of these metrics flow when both Random Forest (RF) and XGBoost regressors are given fewer weeks of historical aggregated data, are provided in Figure 4, Figure 5, and Figure 6.

TABLE I
BENCHMARK MODEL RESULTS USING REFERENCE SOLUTION CODE AND DATASET [15]

Weeks	Model	MAE (mins)	RMSE (mins)	R^2 Score
N/A	Linear Regression	27.98	36.76	0.47
	Random Forest	21.84	30.15	0.64
	XGBoost	21.39	29.68	0.65

TABLE II
REGRESSION MODEL RESULTS ACROSS DIFFERENT AGGREGATED TRAINING BLOCKS

Weeks	Model	MAE (mins)	RMSE (mins)	R^2 Score
16	Random Forest	18.05 $\pm$ 0.65	24.62 $\pm$ 1.27	0.63 $\pm$ 0.03
	XGBoost	18.62 $\pm$ 0.74	25.54 $\pm$ 1.39	0.60 $\pm$ 0.04
14	Random Forest	17.21 $\pm$ 0.54	23.52 $\pm$ 1.15	0.66 $\pm$ 0.03
	XGBoost	17.75 $\pm$ 0.68	24.33 $\pm$ 1.30	0.64 $\pm$ 0.04
12	Random Forest	17.17 $\pm$ 0.55	23.49 $\pm$ 1.16	0.66 $\pm$ 0.03
	XGBoost	17.76 $\pm$ 0.61	24.38 $\pm$ 1.12	0.64 $\pm$ 0.03
10	Random Forest	16.86 $\pm$ 0.53	23.21 $\pm$ 1.15	0.67 $\pm$ 0.03
	XGBoost	17.53 $\pm$ 0.46	24.05 $\pm$ 1.14	0.65 $\pm$ 0.03
8	Random Forest	16.74 $\pm$ 0.49	22.93 $\pm$ 1.05	0.68 $\pm$ 0.03
	XGBoost	17.24 $\pm$ 0.63	23.60 $\pm$ 1.13	0.66 $\pm$ 0.03
6	Random Forest	16.56 $\pm$ 0.51	22.67 $\pm$ 1.04	0.69 $\pm$ 0.03
	XGBoost	17.07 $\pm$ 0.57	23.50 $\pm$ 1.11	0.66 $\pm$ 0.03

The analysis obtained from all six experiments revealed remarkable insights into MRT predictions and how this is affected by the amount of historical data, especially when compared to all benchmark models. This study's controlled environment proved to be highly effective when predicting race times for athletes, even when the regression models were given the longest possible training block data.

Hence, at 16 weeks, the RF and XGBoost models obtained an MAE of 18.05 and 18.62 minutes, respectively, while the benchmark models achieved an MAE of 21.84 and 21.39 minutes. This shows that the models were nearly 4 minutes more accurate, which is highly significant. Also, when compared to the Linear Regression model, they were almost 10 minutes better, as it attained 27.98 minutes for the same metric. This

Model Performance Over Time

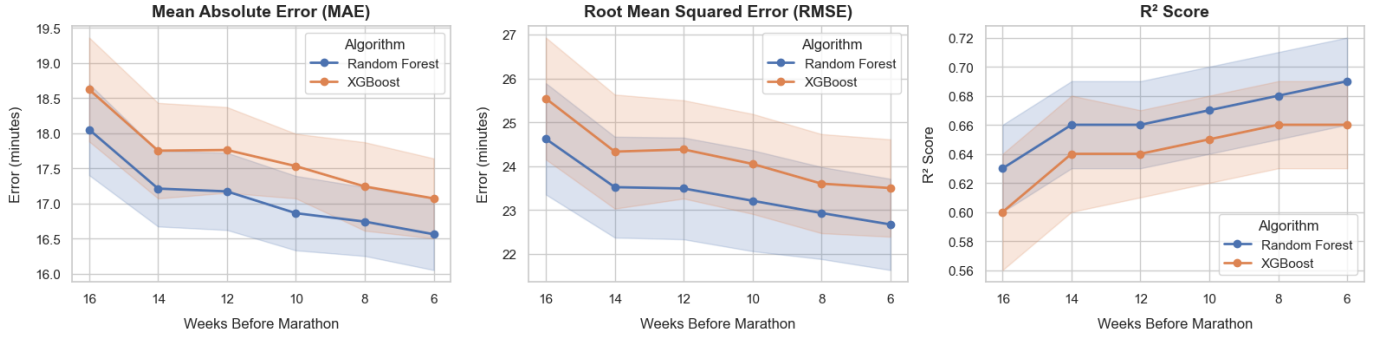


Fig. 4. Regression model predictive performances across different time steps

came after rigorous filtering for the dominant demographic (Males, 35 to 54), precise training block alignment across all athletes in the dataset, and performance category one athlete removal (as they were considered as outliers). In turn, this process eliminated certain noise and provided cleaner data for each model. It could be that the benchmark models encountered difficulties in finding specific patterns between different age groups, genders, and varying training block lengths. This process shows that demographic-specific and time-aligned modelling is crucial in MRT predictions.

When looking at the benchmark models, XGBoost showed higher predictive performance over the RF model (MAE 21.39 against 21.84), while this scenario completely shifted when holding the following controlled experiments, as the RF regression model consistently outperformed XGBoost across all time steps, with Week 6 having an R2 value of 0.69 compared to 0.66, respectively. Having these results demonstrates that a gradient boosting model could fit a heterogeneous and unstructured dataset better than an RF model. In contrast, once the same architecturally structured models were provided with homogenised and training block restrictive datasets, RF showed superiority in understanding the hidden physiological trends without being affected by the noise.

Regarding the temporal aspect of these experiments, both models' predictive power continued to increase linearly as race day got closer. In the case of the RF model, the MAE from Week 16 to Week 6 dropped by 1.49 minutes (18.05 to 16.56). Considering sports science and marathon preparation, running sessions within 6 to 8 weeks of the event provide better predictors for achieving a target performance than those initial training sessions at 16 weeks out. Although there is visible progress by both models within all figures, weeks 14 and 12 depict an anomaly, a training plateau as the MAE marginally shifted, with the R2 value remaining completely static at 0.66 for RF and 0.64 for XGBoost. In marathon training, this period is normally characterised by high training volume rather than intensity, known as 'base building', which results in accumulated fatigue. Such visual and statistical evidence could reflect this stagnation, which makes it difficult for the models to determine a possible MRT.

As shown by the standard deviations in Table II, RF sustained extremely tight error bands, reaching a standard deviation of just ± 0.51 minutes at Week 6. This was due to the 10-fold cross-validation technique that was repeated three times, which makes the results highly reliable and not obtained by coincidence. These values also highlight that an RF regression model is highly robust across various subgroups of the employed demographic. Furthermore, with an MAE of 16.56 minutes and an R2 value of 0.69 at Week 6, this model assists a marathoner with 16.5 minutes above or below the target MRT six weeks from the event, which is realistic and very practical. On the other hand, the R2 value shows that the model managed to explained 69% of the variance in finish times entirely through native, static, and derived training metrics.

Comparing our experimental results with the benchmark models indicates that when the dataset is highly filtered and cleaned, the RF regression model achieves the best predictive performance. This can not be said on a more chaotic dataset, as the XGBoost regressor would possibly be more suitable. In relation to the predictive power, when a regression model is given 16 weeks of aggregated data rather than 6 weeks, it is more prone to predictive mistakes, indicating that recent, race-specific training data is of higher quality for the models than that of earlier stages.

In terms of validity, the methodological approach undertaken, compared to that of the benchmark models, continues to strengthen the results. This was due to rigorous filtering of the dataset to a single demographic, removing outliers based on their performance category within each marathon, and aligning the training periods across all athletes for different experiments, which eliminated substantial confounding variables, such as biological inconsistencies between age groups and genders. Thus, the final MAE, RMSE, and R2 values provide a valid reflection of how marathon preparation is a key aspect of successful prediction, rather than a result of demographic interference. The use of the three-time repeated 10-fold cross-validation technique on identical data splits ensures the reliability of these results. This was also accompanied by shared hyperparameter values across both model architectures.

Essentially, each model was separately trained and tested 30 times on varying slices of data, resulting in a standard deviation, proving consistently stable results. Therefore, it can be confidently stated that the RF regressor is the highest-performing model in this scenario with this dataset and model architecture.

Notably, although the results proved to be successful, accurate, and valid, there is a major loss in generalisability since each model was specifically trained on one demographic from six available ones in the original dataset. The superior model is a specialised tool that could facilitate male runners aged between 35 and 54; apart from that, female runners, and younger or senior male runners will potentially have false and inaccurate results when predicting their MRT. The models were also supplied with data from five of six major marathons due to insufficient activity records. Although applicable to most renowned events, employing them on training data leading to the Tokyo marathon could result in inaccurate results.

This comprehensive analysis leads us to reject the first part of the hypothesis, which stated that an XGBoost regressor would achieve higher predictive performance than an RF regression model. Although the former model is known for its robustness and high accuracy, RF achieved the lowest errors and highest R2 values across all timeframes. The controlled experimental environment and thorough filtering led to such results, and given that XGBoost might have failed to identify certain patterns within this highly specific sample size, while RF is less prone to overfitting. Additionally, the second part of the hypothesis can also be rejected, stating that larger volumes of data (16 weeks) are most accurate when training. Although a wider glance at an athlete's historical data is highly relevant in a training block, the last 6 and 8 weeks of training data proved to yield the best accuracy. This highlights the noise produced by voluminous data sources and the high importance of race-specific metrics closest to race day, which are ultimately more effective.

These results align with the findings of [12], where more complex regression models outperform Linear regressors, making them highly effective for marathon prediction. Also, these results and systematic experiments are supported by [11], where the researchers illustrated that simpler models surpass highly complex ones. Regarding lower errors, [10] also proves this as they state that errors steadily decreased as training progressed and got closer to the marathon event, while having the best model performing on only 6 weeks of aggregated data is reinforced by the methodological approach taken by [12]. Finally, the thorough filtering process held within this study is further reinforced by similar strategic filtering methods held within [12] – [14], as it resulted in significant improvements in model predictions.

V. CONCLUSION

This research investigated the use of tree-based machine learning (ML) regression models in predicting the marathon race time (MRT) of an athlete and evaluated their predictive performance across diverse historical data. Through six

progressive and systematic experiments, this study addressed research questions and a hypothesis, seeking to find a superior model between Random Forest (RF) and Extreme Gradient Boosting (XGBoost) regressors and understand how MRT predictions are affected by timeframe-reduced historical training data.

The main conclusions derived from this study were that simpler tree-based ML regression models, such as the RF regressor, yield more accurate MRT predictions than XGBoost across all time steps, and the RF regression model achieves the best predictive performance when provided with aggregated training block data closer to race day. Once all six experiments were conducted, both research questions were collectively answered. Through the use of three specific metrics, it was observed that both models achieved better predictive results with reduced aggregated training block data. In particular, the RF model showed continuously better results than XGBoost, starting from Week 16 and condensing in 2-week intervals, culminating in peak predictive performance at Week 6. This led to the RF model being the superior one, showing that such models prefer race-day specific data rather than noisy data from the base building stage of a marathon training block. Furthermore, these experiments and their respective conclusions led to rejecting the research hypothesis at the beginning of this study, revealing that XGBoost is not the most accurate ML regressor at Week 16 or subsequent time steps; rather, the contrary is true: the RF model demonstrates consistent superiority in predictive capabilities, reaching its best prediction at Week 6.

While the implemented methodology provided thorough and remarkable findings, some shortcomings should also be noted. As previously mentioned, due to the available dataset, five of the six marathon majors were tested upon since the Tokyo 2019 marathon only had 8.5 weeks of training block data. Moreover, the models were only trained on the most dominant demographic (Males 35 to 54), which makes the regression models less generalisable across different genders and age groups. Also, as discussed in Section III, only two ML regressors were used with a specific, consistent structure and shared hyperparameters.

To address such limitations and further progress research in this domain, numerous possibilities for further investigations are proposed. One suggestion would be to utilise another dataset that captures an entire 16-week training block for the first major of the year, and train the models with this addition. Exploring other specific demographic groups would also be a possible experimental approach, given that such rigorous filtering yielded better results than benchmark models. Additionally, hyperparameter tuning could be applied to such models used within this study to enhance their predictive performance, or even utilise other tree-based ML regressors to see how the results change. These proposed future investigations would provide the running community with actionable insights for race day, and in turn, optimize certain strategies, while more credibility and strength are given to such predictive models in long-distance endurance running.

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